Antioxidant supplements for prevention of gastrointestinal cancers: a systematic review and meta-analysis.

BACKGROUND: Oxidative stress can cause cancer. Our aim was to establish whether antioxidant supplements reduce the incidence of gastrointestinal cancer and mortality. METHODS: With the Cochrane Collaboration methodology, we reviewed all randomised trials comparing antioxidant supplements with placebo for prevention of gastrointestinal cancers. We searched electronic databases and reference lists (February, 2003). Outcome measures were incidence of gastrointestinal cancers, overall mortality, and adverse effects. Outcomes were analysed with fixed-effect and random-effects model meta-analyses and were reported as relative risk with 95% CIs. FINDINGS: We identified 14 randomised trials (n=170,525). Trial quality was generally high. Heterogeneity of results was low to moderate. Neither the fixed-effect (relative risk 0.96, 95% CI 0.88-1.04) nor random-effects meta-analyses (0.90, 0.77-1.05) showed significant effects of supplementation with beta-carotene, vitamins A, C, E, and selenium (alone or in combination) versus placebo on oesophageal, gastric, colorectal, pancreatic, and liver cancer incidences. In seven high-quality trials (n=131727), the fixed-effect model showed that antioxidant significantly increased mortality (1.06, 1.02-1.10), unlike the random-effects meta-analysis (1.06, 0.98-1.15). Low-quality trials showed no significant effect of antioxidant supplementation on mortality. The difference between the mortality estimates in high-quality and low-quality trials was significant (Z=2.10, p=0.04 by test of interaction). beta-carotene and vitamin A (1.29, 1.14-1.45) and beta-carotene and vitamin E (1.10, 1.01-1.20) significantly increased mortality, whereas beta-carotene alone only tended to increase mortality (1.05, 0.99-1.11). In four trials (three with unclear or inadequate methodology), selenium showed significant beneficial effect on the incidence of gastrointestinal cancer. INTERPRETATION: We could not find evidence that antioxidant supplements can prevent gastrointestinal cancers; on the contrary, they seem to increase overall mortality. The potential preventive effect of selenium should be studied in adequate randomised trials.